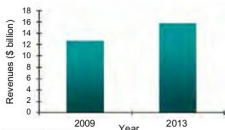


Microcontrollers Market: Revenue Forecasts (World), 2009 and 2013



Source: Frost & Sullivan

evolution of Flash technologies, higher performance and higher integration.

Since microcontrollers are, for the most part, a system-on-a-chip, the technology changes are typically incremental in the various peripherals. For example, there is an ever-increasing demand for connectivity options like USB, Ethernet, CAN and wireless. This connectivity requires not only hardware changes but also the associated software libraries and development tools to integrate it within an application.

Move from 16-bit to 32-bit. Applications like industrial automation, medical equipment, point-of-sale terminals, digital weighing scales, printers, and motor control and display have migrated from 8-bit/16-bit MCUs to 32-bit platform as 32-bit MCUs are now available at the price of 8-bit MCUs. Some of the key advantages of 32-bit MCUs are high operating frequency, low active-power requirement, code reuse, code size reduction by 40-50 per cent, enhanced features for increasingly complex algorithms and availability of tools.

ARM Cortex-M. The major market trend has been the use of more and more third-party embedded cores. "Proprietary cores may start phasing out slowly or will not evolve further with the growing use of embedded cores like ARM-core," says Vinay Thapliyal, technical marketing manager, MCU division, STMicroelectronics.

According to Semicast Research, ARM would become the leading 32-bit microcontroller architecture in 2011, ahead of Power Architecture and x86.

ARM Cortex-M series, where 'M'

stands for microcontrollers, has dominant share in 32-bit MCU offerings. The different variants like Cortex-M0, M3 and most recent M4 serve the speed range from 50 MHz up to 150 MHz. The ARM architecture continues to gain momentum in the MCU market, with vendors like Atmel, Texas Instruments, NXP Semiconductors and STMicro-

electronics all demonstrating Cortex-M processor-based MCUs.

On-chip memory. Evolution of integrated program memory from ROM to Flash has greatly influenced MCUs. Today, flash-based MCUs are widely used in real-time control applications.

"The biggest benefit of on-chip flash memories is that these significantly reduce the cost of production and expansion of real-time adaptive control applications, providing a cost/value advantage. Acceptance of Flash as the standard for MCUs is one of the biggest technology trends," says Andreas Eieland, AVR product marketing manager, Atmel.

Thapliyal informs, "Today, we have devices up to 1 MB of Flash and more than 100 kB of SRAM to serve the increasing demand for software tools and applications that demand higher integration of the on-chip memory."

Power efficiency. Fanie Duvenhage, director of applications, architecture and marketing security, microcontroller and technology development division, Microchip, says, "Over the last few years, lower power consumption has become more important with the current drive towards energy efficiency leading to introduction of ultra-low-power microcontrollers and device power management solutions. Low-current sleep modes, low standby power and auto switching when idle are some of the common power management features seen in MCUs.

One of the strongest technology trends is that many vendors are introducing products with multiple internal voltage domains to reduce active-mode

Vendor profiles

Fujitsu

Fujitsu microcontrollers are widely used in automobile, home, industrial and consumer applications. They include general-purpose 8-, 16-, 32-bit MCUs, and specialised ICs for peripheral control. A variety of development kits, in-circuit emulators, productivity-boosting tools, and comprehensive support help Fujitsu's customers accelerate development of their software. Fujitsu microcontroller families include 8-bit-FMC/8FX, 16-bit-FMC-16LX and FMC-16FX and 32-bit-FR.

Infineon

Infineon's major focus areas are energy efficiency, mobility and security. To cover these energy-efficient markets, Infineon has a complete portfolio of 8-/16-/32-bit microcontrollers.

Infineon's 8-bit controller portfolio features application-specific peripherals like CCU6, fast ADC, USIC (universal serial interface controller) and CAN (controller area network). These are completely scalable from 16-pin 2k flash to 64-pin 64k flash with complete code compatibility. In 16-bit MCUs, scalability is available from 64-pin 64k flash to 176-pin 1.76MB flash. For consumer market applications like washing machines, household fans and refrigerators, Infineon's 8- or 16-bit portfolio products are suitable.

For telecom, automotive and high-end industrial segments, Infineon has a very strong portfolio of 32-bit controllers that can range from 144-pin 1.5MB to 576-pin 5.6MB flash.

Microchip

Microchip Technology manufactures the popular PIC microcontrollers (MCUs), with more than 700 different products covering a broad range of 8-, 16- and 32-bit MCUs and digital signal controllers (DSCs). These cover the general-purpose MCU market, along with various specialty MCU peripherals to address such applications as motor control, touch sensing, graphics, USB and Ethernet switch-mode power supplies. All the microcontrollers are supported by Microchip's MPLAB integrated development environment.

power consumption. Eieland shares that vendors are providing several different solutions to overcome the challenge to maintain RAM retention and have a short wake-up time from sleep without too much leakage current.

Low-power microcontrollers are in